Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Circumference of a circle of radius 1?
2. How many degrees is a quarter turn?
3. Calculate $\sqrt{\frac{1}{2}}$.
4. What are the x- and y-coordinates of the reflection of (a, b) in the x-axis?
5. Simplify $\frac{5\pi}{3} - 2\pi$.
6. In a right triangle, opposite side 3 and hypotenuse 5: sine?

After the diagnostic. Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

Speaking and logic. Present a trigonometric identity involving related angles, naming the symmetry that justifies it; explain what a radian measures.

Investigate before formalising

Construct, without a protractor, a point at angle $\frac{\pi}{3}$ then a point at angle $\frac{\pi}{4}$. Justify the exact values of their coordinates from the figures.

Organisation. Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.